EduPath — Final Database Design

Final schema reflecting the team's agreed changes • 8 tables • MySQL

Overview

This document lists the final tables, columns, data types, key constraints, responsible team member, and consuming module. The questions table does not include guide_post_id. Alumni details are stored in alumni_profiles.

1. Users and Roles

Table: users

Column	Type	Key / Notes
user_id	BIGINT	Primary key; auto-increment
reg_no	VARCHAR(30)	Unique; nullable if not applicable
name	VARCHAR(100)	Required
email	VARCHAR(150)	Unique; required
hashed_password	VARCHAR(255)	Required; store a secure password hash
role	VARCHAR(20)	Required; JUNIOR / SENIOR / ALUMNI / ADMIN

Responsible: Person 1. **Consumed by:** all modules for authentication, account identity, and role-based permissions.

2. Alumni Profiles

Table: alumni_profiles

Column	Type	Key / Notes
alumni_id	BIGINT	Primary key and foreign key to users.user_id
current_company	VARCHAR(150)	Optional; company where the alumnus currently works
job_title	VARCHAR(150)	Optional; current professional role

Relationship: One users row can have at most one alumni_profiles row; alumni_id reuses the corresponding user_id.
Responsible: Person 1 / team database integration. **Consumed by:** alumni profile and mentorship features.

3. Senior / Alumni Guidance

Table: playbook_posts

Column	Type	Key / Notes
post_id	BIGINT	Primary key; auto-increment
author_id	BIGINT	Required; foreign key to users.user_id
title	VARCHAR(200)	Required
markdown_content	TEXT	Guidance content
category	VARCHAR(40)	Academics / Internships / Higher Studies / Placement
status	VARCHAR(20)	DRAFT / PUBLISHED
created_at	TIMESTAMP	Defaults to current timestamp

Responsible: Person 3. **Consumed by:** senior/alumni guidance and the junior feed. The application enforces whether SENIOR and/or ALUMNI users may publish.

4. Questions and Answers

Table: questions

Column	Type	Key / Notes
question_id	BIGINT	Primary key; auto-increment
author_id	BIGINT	Required; foreign key to users.user_id
title	VARCHAR(200)	Required; question title
question_body	TEXT	Required; question text
category	VARCHAR(40)	Academics / Internships / Higher Studies / Placement
created_at	TIMESTAMP	Defaults to current timestamp

Note: guide_post_id is intentionally omitted. **Responsible:** Person 2. **Consumed by:** Q&A module and junior feed.

Table: answers

Column	Type	Key / Notes
answer_id	BIGINT	Primary key; auto-increment
question_id	BIGINT	Required; foreign key to questions.question_id
author_id	BIGINT	Required; foreign key to users.user_id
answer_body	TEXT	Required; answer text
is_verified_answer	BOOLEAN	Defaults to FALSE
created_at	TIMESTAMP	Defaults to current timestamp

Responsible: Person 2. **Consumed by:** Q&A module; verified-answer state supports marking an accepted/verified response.

5. AI Career Roadmaps

Table: roadmaps

Column	Type	Key / Notes
roadmap_id	BIGINT	Primary key; auto-increment
user_id	BIGINT	Required; foreign key to users.user_id; roadmap owner
goal	VARCHAR(255)	Student's career goal
title	VARCHAR(200)	Generated roadmap title
created_at	TIMESTAMP	Defaults to current timestamp

Table: roadmap_nodes

Column	Type	Key / Notes
node_id	BIGINT	Primary key; auto-increment
roadmap_id	BIGINT	Required; foreign key to roadmaps.roadmap_id
title	VARCHAR(200)	Required; skill or milestone title
description	TEXT	What to learn or do
semester	INT	Optional; suggested semester
sequence_order	INT	Display order

Table: roadmap_edges

Column	Type	Key / Notes
source_node_id	BIGINT	Part of composite primary key; foreign key to roadmap_nodes.node_id; prerequisite node
target_node_id	BIGINT	Part of composite primary key; foreign key to roadmap_nodes.node_id; dependent node

Primary key: roadmap_edges uses the composite key (source_node_id, target_node_id), preventing duplicate directed edges. **Responsible:** Person 4 owns roadmap node/graph entities and graph API; Person 5 generates roadmap content with Spring AI/Gemini and saves it. **Consumed by:** AI career path and roadmap visualization modules.

6. Relationship Summary

Relationship	Cardinality / Purpose
users → alumni_profiles	One-to-zero-or-one; alumni profile reuses users.user_id
users → playbook_posts	One-to-many; each post has one author
users → questions	One-to-many; each question has one author
questions → answers	One-to-many; each answer belongs to one question
users → answers	One-to-many; each answer has one author
users → roadmaps	One-to-many; each roadmap has one owner
roadmaps → roadmap_nodes	One-to-many; each node belongs to one roadmap
roadmap_nodes → roadmap_edges	Each edge references a source node and a target node